XLVII National Mathematics Olympiad: 3rd round, 25–26 April 1998

Problem 1. Find the least positive integer number n $(n \geq 3)$ with the following property: for any colouring of n different points $A_1, A_2, \ldots, A_n$ on a line and such that $A_1A_2 = A_2A_3 = \ldots = A_{n-1}A_n$ in two colours, there are three points $A_i, A_j, A_{2j-i} (1 \leq i < 2j - i \leq n)$ which have the same colour.

Problem 2. Let $ABCD$ be a quadrilateral such that $AD = CD$ and $\angle DAB = \angle ABC < 90°$. The line passing through D and the midpoint of the segment BC intersects the line AB at the point E. Prove that $\angle BEC = \angle DAC$.

Problem 3. Let $\mathbb{R}^+$ be the set of all positive real numbers. Prove that there is no function $f : \mathbb{R}^+ \to \mathbb{R}^+$ such that

$$(f(x))^2 \geq f(x + y)(f(x) + y)$$

for arbitrary positive real numbers x and y.

Problem 4. Let $f(x) = x^3 - 3x + 1$. Find the number of different real solutions of the equation $f(f(x)) = 0$.

Problem 5. The convex pentagon $ABCDE$ is inscribed in a circle with radius R. The inradii of the triangles ABC, ABD, AEC and AED are denoted by r_{ABC}, r_{ABD}, r_{AEC} and r_{AED}. Prove that

a.) $\cos \angle CAB + \cos \angle ABC + \cos \angle BCA = 1 + \dfrac{r_{ABC}}{R}$;

b.) If $r_{ABC} = r_{AED}$ and $r_{ABD} = r_{AEC}$, then $\triangle ABC \cong \triangle AED$.

Problem 6. Show that the equation

$$x^2y^2 = z^2(z^2 - x^2 - y^2)$$

has no solution in positive integer numbers.

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